In [1]:
```
!pip install langchain==0.3.27
!pip install langchain-core==0.3.72
!pip install langchain-community==0.3.27
# Suuportive packages for Langchain to read PDF Files.
!pip install pypdf
# Supportive packages for Langchain to read DOCX, TXTFiles.
!pip install tiktoken
# Install ChromaDB and Faiss for vector storage and similarity search
!pip install chromadb
!pip install faiss-cpu
!pip install google-genai
```

```
Requirement already satisfied: langchain==0.3.27 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (0.3.27)
Requirement already satisfied: langchain-core<1.0.0,>=0.3.72 in C:\Users\shreya.
2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langcha
in==0.3.27) (0.3.72)
Requirement already satisfied: langchain-text-splitters<1.0.0,>=0.3.9 in C:\Users
\shreya.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
langchain==0.3.27) (0.3.9)
Requirement already satisfied: langsmith>=0.1.17 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from langchain==0.3.27)
(0.14.1)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain==0.3.
27) (2.12.5)
Requirement already satisfied: SQLAlchemy<3,>=1.4 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from langchain==0.3.27)
(2.1.1)
Requirement already satisfied: requests<3,>=2 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from langchain==0.3.27) (2.3
4.2)
Requirement already satisfied: PyYAML>=5.3 in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from langchain==0.3.27) (6.0.3)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in C:\Users\shrey
a.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langc
hain-core<1.0.0,>=0.3.72->langchain==0.3.27) (9.1.4)
Requirement already satisfied: jsonpatch<2.0,>=1.33 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<
1.0.0,>=0.3.72->langchain==0.3.27) (1.33)
Requirement already satisfied: typing-extensions>=4.7 in C:\Users\shreya.2.agrawa
l\AppData\Roaming\Python\Python313\site-packages (from langchain-core<1.0.0,>=0.
3.72->langchain==0.3.27) (4.16.0)
Requirement already satisfied: packaging>=23.2 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<1.0.0,>
=0.3.72->langchain==0.3.27) (25.0)
Requirement already satisfied: jsonpointer>=1.9 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from jsonpatch<2.0,>=1.33-
>langchain-core<1.0.0,>=0.3.72->langchain==0.3.27) (3.1.1)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3.0.0,
>=2.7.4->langchain==0.3.27) (0.8.0)
Requirement already satisfied: pydantic-core==2.41.5 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3.0.0,>
=2.7.4->langchain==0.3.27) (2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3.0.
0,>=2.7.4->langchain==0.3.27) (0.4.4)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from requests<3,>=
2->langchain==0.3.27) (3.5.1)
Requirement already satisfied: idna<4,>=2.5 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from requests<3,>=2->langchai
n==0.3.27) (3.20)
Requirement already satisfied: urllib3<3,>=1.26 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from requests<3,>=2->langc
hain==0.3.27) (2.8.0)
Requirement already satisfied: certifi>=2023.5.7 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from requests<3,>=2->lang
chain==0.3.27) (2026.7.22)
Requirement already satisfied: anyio>=3.5.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.17->langc
```

```
hain==0.3.27) (4.15.1)
Requirement already satisfied: distro>=1.7.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.17->langc
hain==0.3.27) (1.9.0)
Requirement already satisfied: httpx2<3,>=2 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.17->langc
hain==0.3.27) (2.13.1)
Requirement already satisfied: orjson>=3.9.14 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.17->lang
chain==0.3.27) (3.12.0)
Requirement already satisfied: requests-toolbelt>=1.0.0 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.
1.17->langchain==0.3.27) (1.0.0)
Requirement already satisfied: sniffio>=1.1 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.17->langc
hain==0.3.27) (1.3.1)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.
1.17->langchain==0.3.27) (0.17.1)
Requirement already satisfied: websockets>=15.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.17->la
ngchain==0.3.27) (16.1.1)
Requirement already satisfied: xxhash>=3.0.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.17->langc
hain==0.3.27) (4.0.1)
Requirement already satisfied: zstandard>=0.23.0 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.17->l
angchain==0.3.27) (0.25.0)
Requirement already satisfied: httpcore2==2.13.1 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsm
ith>=0.1.17->langchain==0.3.27) (2.13.1)
Requirement already satisfied: truststore>=0.10 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsmi
th>=0.1.17->langchain==0.3.27) (0.10.4)
Requirement already satisfied: h11>=0.16 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from httpcore2==2.13.1->httpx2<3,
>=2->langsmith>=0.1.17->langchain==0.3.27) (0.16.0)
Requirement already satisfied: langchain-core==0.3.72 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (0.3.72)
Requirement already satisfied: langsmith>=0.3.45 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core==0.3.
72) (0.14.1)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in C:\Users\shrey
a.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langc
hain-core==0.3.72) (9.1.4)
Requirement already satisfied: jsonpatch<2.0,>=1.33 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core==
0.3.72) (1.33)
Requirement already satisfied: PyYAML>=5.3 in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from langchain-core==0.3.72)
(6.0.3)
Requirement already satisfied: typing-extensions>=4.7 in C:\Users\shreya.2.agrawa
l\AppData\Roaming\Python\Python313\site-packages (from langchain-core==0.3.72)
(4.16.0)
Requirement already satisfied: packaging>=23.2 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core==0.3.7
2) (25.0)
Requirement already satisfied: pydantic>=2.7.4 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core==0.3.7
2) (2.12.5)
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Requirement already satisfied: jsonpointer>=1.9 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from jsonpatch<2.0,>=1.33-
>langchain-core==0.3.72) (3.1.1)
Requirement already satisfied: anyio>=3.5.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core==0.3.72) (4.15.1)
Requirement already satisfied: distro>=1.7.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core==0.3.72) (1.9.0)
Requirement already satisfied: httpx2<3,>=2 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core==0.3.72) (2.13.1)
Requirement already satisfied: orjson>=3.9.14 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->lang
chain-core==0.3.72) (3.12.0)
Requirement already satisfied: requests-toolbelt>=1.0.0 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.
3.45->langchain-core==0.3.72) (1.0.0)
Requirement already satisfied: requests>=2.0.0 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->lan
gchain-core==0.3.72) (2.34.2)
Requirement already satisfied: sniffio>=1.1 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core==0.3.72) (1.3.1)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.
3.45->langchain-core==0.3.72) (0.17.1)
Requirement already satisfied: websockets>=15.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->la
ngchain-core==0.3.72) (16.1.1)
Requirement already satisfied: xxhash>=3.0.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core==0.3.72) (4.0.1)
Requirement already satisfied: zstandard>=0.23.0 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->l
angchain-core==0.3.72) (0.25.0)
Requirement already satisfied: httpcore2==2.13.1 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsm
ith>=0.3.45->langchain-core==0.3.72) (2.13.1)
Requirement already satisfied: idna>=3.18 in C:\Users\shreya.2.agrawal\AppData\Lo
cal\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsmith>=0.
3.45->langchain-core==0.3.72) (3.20)
Requirement already satisfied: truststore>=0.10 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsmi
th>=0.3.45->langchain-core==0.3.72) (0.10.4)
Requirement already satisfied: h11>=0.16 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from httpcore2==2.13.1->httpx2<3,
>=2->langsmith>=0.3.45->langchain-core==0.3.72) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic>=2.7.4
->langchain-core==0.3.72) (0.8.0)
Requirement already satisfied: pydantic-core==2.41.5 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic>=2.7.4-
>langchain-core==0.3.72) (2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic>=2.
7.4->langchain-core==0.3.72) (0.4.4)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from requests>=2.
0.0->langsmith>=0.3.45->langchain-core==0.3.72) (3.5.1)
```

```
Requirement already satisfied: urllib3<3,>=1.26 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from requests>=2.0.0->lang
smith>=0.3.45->langchain-core==0.3.72) (2.8.0)
Requirement already satisfied: certifi>=2023.5.7 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from requests>=2.0.0->lan
gsmith>=0.3.45->langchain-core==0.3.72) (2026.7.22)
Requirement already satisfied: langchain-community==0.3.27 in C:\Users\shreya.2.a
grawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (0.3.27)
Requirement already satisfied: langchain-core<1.0.0,>=0.3.66 in C:\Users\shreya.
2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langcha
in-community==0.3.27) (0.3.72)
Requirement already satisfied: langchain<1.0.0,>=0.3.26 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-com
munity==0.3.27) (0.3.27)
Requirement already satisfied: SQLAlchemy<3,>=1.4 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-community
==0.3.27) (2.1.1)
Requirement already satisfied: requests<3,>=2 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from langchain-community==0.
3.27) (2.34.2)
Requirement already satisfied: PyYAML>=5.3 in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from langchain-community==0.3.2
7) (6.0.3)
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-commun
ity==0.3.27) (3.14.3)
Requirement already satisfied: tenacity!=8.4.0,<10,>=8.1.0 in C:\Users\shreya.2.a
grawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-
community==0.3.27) (9.1.4)
Requirement already satisfied: dataclasses-json<0.7,>=0.5.7 in C:\Users\shreya.2.
agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain
-community==0.3.27) (0.6.7)
Requirement already satisfied: pydantic-settings<3.0.0,>=2.4.0 in C:\Users\shrey
a.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langc
hain-community==0.3.27) (2.15.0)
Requirement already satisfied: langsmith>=0.1.125 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-community
==0.3.27) (0.14.1)
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-comm
unity==0.3.27) (0.4.3)
Requirement already satisfied: numpy>=2.1.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langchain-community==0.
3.27) (2.4.4)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,
>=3.8.3->langchain-community==0.3.27) (2.7.1)
Requirement already satisfied: aiosignal>=1.4.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.8.3
->langchain-community==0.3.27) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.8.3->l
angchain-community==0.3.27) (26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.8.
3->langchain-community==0.3.27) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.
8.3->langchain-community==0.3.27) (6.9.1)
Requirement already satisfied: propcache>=0.2.0 in C:\Users\shreya.2.agrawal\AppD
```

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ata\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.8.3
->langchain-community==0.3.27) (0.5.4)
Requirement already satisfied: yarl<2.0,>=1.17.0 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.8.
3->langchain-community==0.3.27) (1.25.1)
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in C:\Users\shreya.2.ag
rawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from dataclasses
-json<0.7,>=0.5.7->langchain-community==0.3.27) (3.26.2)
Requirement already satisfied: typing-inspect<1,>=0.4.0 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from dataclasses-j
son<0.7,>=0.5.7->langchain-community==0.3.27) (0.9.0)
Requirement already satisfied: langchain-text-splitters<1.0.0,>=0.3.9 in C:\Users
\shreya.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
langchain<1.0.0,>=0.3.26->langchain-community==0.3.27) (0.3.9)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain<1.0.
0,>=0.3.26->langchain-community==0.3.27) (2.12.5)
Requirement already satisfied: jsonpatch<2.0,>=1.33 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<
1.0.0,>=0.3.66->langchain-community==0.3.27) (1.33)
Requirement already satisfied: typing-extensions>=4.7 in C:\Users\shreya.2.agrawa
l\AppData\Roaming\Python\Python313\site-packages (from langchain-core<1.0.0,>=0.
3.66->langchain-community==0.3.27) (4.16.0)
Requirement already satisfied: packaging>=23.2 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<1.0.0,>
=0.3.66->langchain-community==0.3.27) (25.0)
Requirement already satisfied: jsonpointer>=1.9 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from jsonpatch<2.0,>=1.33-
>langchain-core<1.0.0,>=0.3.66->langchain-community==0.3.27) (3.1.1)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3.0.0,
>=2.7.4->langchain<1.0.0,>=0.3.26->langchain-community==0.3.27) (0.8.0)
Requirement already satisfied: pydantic-core==2.41.5 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3.0.0,>
=2.7.4->langchain<1.0.0,>=0.3.26->langchain-community==0.3.27) (2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3.0.
0,>=2.7.4->langchain<1.0.0,>=0.3.26->langchain-community==0.3.27) (0.4.4)
Requirement already satisfied: python-dotenv>=0.21.0 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic-setting
s<3.0.0,>=2.4.0->langchain-community==0.3.27) (1.2.3)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from requests<3,>=
2->langchain-community==0.3.27) (3.5.1)
Requirement already satisfied: idna<4,>=2.5 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from requests<3,>=2->langchai
n-community==0.3.27) (3.20)
Requirement already satisfied: urllib3<3,>=1.26 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from requests<3,>=2->langc
hain-community==0.3.27) (2.8.0)
Requirement already satisfied: certifi>=2023.5.7 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from requests<3,>=2->lang
chain-community==0.3.27) (2026.7.22)
Requirement already satisfied: mypy-extensions>=0.3.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from typing-inspect<
1,>=0.4.0->dataclasses-json<0.7,>=0.5.7->langchain-community==0.3.27) (1.1.0)
Requirement already satisfied: anyio>=3.5.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.125->lang
chain-community==0.3.27) (4.15.1)
Requirement already satisfied: distro>=1.7.0 in C:\Users\shreya.2.agrawal\AppData
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\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.125->lang
chain-community==0.3.27) (1.9.0)
Requirement already satisfied: httpx2<3,>=2 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.125->lang
chain-community==0.3.27) (2.13.1)
Requirement already satisfied: orjson>=3.9.14 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.125->lan
gchain-community==0.3.27) (3.12.0)
Requirement already satisfied: requests-toolbelt>=1.0.0 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.
1.125->langchain-community==0.3.27) (1.0.0)
Requirement already satisfied: sniffio>=1.1 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.125->lang
chain-community==0.3.27) (1.3.1)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.
1.125->langchain-community==0.3.27) (0.17.1)
Requirement already satisfied: websockets>=15.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.125->l
angchain-community==0.3.27) (16.1.1)
Requirement already satisfied: xxhash>=3.0.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.125->lang
chain-community==0.3.27) (4.0.1)
Requirement already satisfied: zstandard>=0.23.0 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.1.125->
langchain-community==0.3.27) (0.25.0)
Requirement already satisfied: httpcore2==2.13.1 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsm
ith>=0.1.125->langchain-community==0.3.27) (2.13.1)
Requirement already satisfied: truststore>=0.10 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsmi
th>=0.1.125->langchain-community==0.3.27) (0.10.4)
Requirement already satisfied: h11>=0.16 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from httpcore2==2.13.1->httpx2<3,
>=2->langsmith>=0.1.125->langchain-community==0.3.27) (0.16.0)
Requirement already satisfied: pypdf in C:\Users\shreya.2.agrawal\AppData\Local\P
rograms\Python\Python313\Lib\site-packages (6.19.0)
Requirement already satisfied: tiktoken in C:\Users\shreya.2.agrawal\AppData\Loca
l\Programs\Python\Python313\Lib\site-packages (0.14.0)
Requirement already satisfied: regex in C:\Users\shreya.2.agrawal\AppData\Local\P
rograms\Python\Python313\Lib\site-packages (from tiktoken) (2026.9.29)
Requirement already satisfied: requests in C:\Users\shreya.2.agrawal\AppData\Loca
l\Programs\Python\Python313\Lib\site-packages (from tiktoken) (2.34.2)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from requests->tik
token) (3.5.1)
Requirement already satisfied: idna<4,>=2.5 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from requests->tiktoken) (3.2
0)
Requirement already satisfied: urllib3<3,>=1.26 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from requests->tiktoken)
(2.8.0)
Requirement already satisfied: certifi>=2023.5.7 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from requests->tiktoken)
(2026.7.22)
Requirement already satisfied: chromadb in C:\Users\shreya.2.agrawal\AppData\Loca
l\Programs\Python\Python313\Lib\site-packages (1.5.9)
Requirement already satisfied: build>=1.0.3 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (1.6.1)
Requirement already satisfied: pydantic>=2.0 in C:\Users\shreya.2.agrawal\AppData
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\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (2.12.5)
Requirement already satisfied: pydantic-settings>=2.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (2.1
5.0)
Requirement already satisfied: pybase64>=1.4.1 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (1.5.0)
Requirement already satisfied: uvicorn>=0.18.3 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from uvicorn[standard]>=0.1
8.3->chromadb) (0.35.0)
Requirement already satisfied: numpy>=1.22.5 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (2.4.4)
Requirement already satisfied: typing-extensions>=4.5.0 in C:\Users\shreya.2.agra
wal\AppData\Roaming\Python\Python313\site-packages (from chromadb) (4.16.0)
Requirement already satisfied: onnxruntime>=1.14.1 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (1.30.0)
Requirement already satisfied: opentelemetry-api>=1.2.0 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (1.
45.0)
Requirement already satisfied: opentelemetry-exporter-otlp-proto-grpc>=1.2.0 in
C:\Users\shreya.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packag
es (from chromadb) (1.45.0)
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (1.
45.0)
Requirement already satisfied: tokenizers>=0.13.2 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (0.23.2)
Requirement already satisfied: pypika>=0.48.9 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (0.51.1)
Requirement already satisfied: tqdm>=4.65.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (4.70.1)
Requirement already satisfied: overrides>=7.3.1 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (7.7.0)
Requirement already satisfied: importlib-resources in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (7.1.0)
Requirement already satisfied: grpcio>=1.58.0 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (1.84.0)
Requirement already satisfied: bcrypt>=4.0.1 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (5.0.0)
Requirement already satisfied: typer>=0.9.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (0.27.2)
Requirement already satisfied: kubernetes>=28.1.0 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (36.0.3)
Requirement already satisfied: tenacity>=8.2.3 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (9.1.4)
Requirement already satisfied: pyyaml>=6.0.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (6.0.3)
Requirement already satisfied: mmh3>=4.0.1 in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from chromadb) (5.3.0)
Requirement already satisfied: orjson>=3.9.12 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (3.12.0)
Requirement already satisfied: httpx>=0.27.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (15.0.0)
Requirement already satisfied: jsonschema>=4.19.0 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from chromadb) (4.26.0)
Requirement already satisfied: packaging>=24.0 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from build>=1.0.3->chromad
b) (25.0)
Requirement already satisfied: pyproject_hooks in C:\Users\shreya.2.agrawal\AppDa
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ta\Local\Programs\Python\Python313\Lib\site-packages (from build>=1.0.3->chromad
b) (1.3.3)
Requirement already satisfied: colorama in C:\Users\shreya.2.agrawal\AppData\Roam
ing\Python\Python313\site-packages (from build>=1.0.3->chromadb) (0.4.6)
Requirement already satisfied: anyio in C:\Users\shreya.2.agrawal\AppData\Local\P
rograms\Python\Python313\Lib\site-packages (from httpx>=0.27.0->chromadb) (4.15.
1)
Requirement already satisfied: certifi in C:\Users\shreya.2.agrawal\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from httpx>=0.27.0->chromadb) (202
6.7.22)
Requirement already satisfied: httpcore==1.* in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from httpx>=0.27.0->chromadb)
(1.0.9)
Requirement already satisfied: idna in C:\Users\shreya.2.agrawal\AppData\Local\Pr
ograms\Python\Python313\Lib\site-packages (from httpx>=0.27.0->chromadb) (3.20)
Requirement already satisfied: h11>=0.16 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from httpcore==1.*->httpx>=0.27.0
->chromadb) (0.16.0)
Requirement already satisfied: attrs>=22.2.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from jsonschema>=4.19.0->chro
madb) (26.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in C:\Users\s
hreya.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from j
sonschema>=4.19.0->chromadb) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from jsonschema>=4.19.0
->chromadb) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from jsonschema>=4.19.0->ch
romadb) (2026.6.3)
Requirement already satisfied: six>=1.9.0 in C:\Users\shreya.2.agrawal\AppData\Ro
aming\Python\Python313\site-packages (from kubernetes>=28.1.0->chromadb) (1.17.0)
Requirement already satisfied: python-dateutil>=2.5.3 in C:\Users\shreya.2.agrawa
l\AppData\Roaming\Python\Python313\site-packages (from kubernetes>=28.1.0->chroma
db) (2.9.0.post0)
Requirement already satisfied: websocket-client!=0.40.0,!=0.41.*,!=0.42.*,>=0.32.
0 in C:\Users\shreya.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-p
ackages (from kubernetes>=28.1.0->chromadb) (1.9.2)
Requirement already satisfied: requests in C:\Users\shreya.2.agrawal\AppData\Loca
l\Programs\Python\Python313\Lib\site-packages (from kubernetes>=28.1.0->chromadb)
(2.34.2)
Requirement already satisfied: requests-oauthlib in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from kubernetes>=28.1.0->
chromadb) (2.0.0)
Requirement already satisfied: urllib3!=2.6.0,>=1.24.2 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from kubernetes>=2
8.1.0->chromadb) (2.8.0)
Requirement already satisfied: durationpy>=0.7 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from kubernetes>=28.1.0->ch
romadb) (0.11)
Requirement already satisfied: aiohttp<4.0.0,>=3.13.5 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from kubernetes>=28.
1.0->chromadb) (3.14.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,
>=3.13.5->kubernetes>=28.1.0->chromadb) (2.7.1)
Requirement already satisfied: aiosignal>=1.4.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.13.
5->kubernetes>=28.1.0->chromadb) (1.4.0)
Requirement already satisfied: frozenlist>=1.1.1 in C:\Users\shreya.2.agrawal\App
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Data\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.1
3.5->kubernetes>=28.1.0->chromadb) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.
13.5->kubernetes>=28.1.0->chromadb) (6.9.1)
Requirement already satisfied: propcache>=0.2.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.13.
5->kubernetes>=28.1.0->chromadb) (0.5.4)
Requirement already satisfied: yarl<2.0,>=1.17.0 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.1
3.5->kubernetes>=28.1.0->chromadb) (1.25.1)
Requirement already satisfied: flatbuffers in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from onnxruntime>=1.14.1->chrom
adb) (25.12.19)
Requirement already satisfied: protobuf>=4.25.8 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from onnxruntime>=1.14.1->
chromadb) (4.25.9)
Requirement already satisfied: googleapis-common-protos~=1.57 in C:\Users\shreya.
2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from opentel
emetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.75.0)
Requirement already satisfied: opentelemetry-exporter-otlp-common==0.66b0 in C:\U
sers\shreya.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages
(from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (0.66b0)
Requirement already satisfied: opentelemetry-exporter-otlp-proto-common==1.45.0 i
n C:\Users\shreya.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-pack
ages (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.45.0)
Requirement already satisfied: opentelemetry-proto==1.45.0 in C:\Users\shreya.2.a
grawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from openteleme
try-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.45.0)
Collecting protobuf>=4.25.8 (from onnxruntime>=1.14.1->chromadb)
  Using cached protobuf-7.36.2-cp310-abi3-win_amd64.whl.metadata (595 bytes)
Requirement already satisfied: opentelemetry-semantic-conventions==0.66b0 in C:\U
sers\shreya.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages
(from opentelemetry-sdk>=1.2.0->chromadb) (0.66b0)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic>=2.0->
chromadb) (0.8.0)
Requirement already satisfied: pydantic-core==2.41.5 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic>=2.0->c
hromadb) (2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic>=2.0
->chromadb) (0.4.4)
Requirement already satisfied: python-dotenv>=0.21.0 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic-setting
s>=2.0->chromadb) (1.2.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from rich>=10.11.0->c
hromadb) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in C:\Users\shreya.2.agraw
al\AppData\Roaming\Python\Python313\site-packages (from rich>=10.11.0->chromadb)
(2.21.0)
Requirement already satisfied: mdurl~=0.1 in C:\Users\shreya.2.agrawal\AppData\Lo
cal\Programs\Python\Python313\Lib\site-packages (from markdown-it-py>=2.2.0->rich
>=10.11.0->chromadb) (0.1.2)
Requirement already satisfied: huggingface-hub<2.0,>=0.16.4 in C:\Users\shreya.2.
agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from tokenizer
s>=0.13.2->chromadb) (1.33.0)
Requirement already satisfied: click<9.0.0,>=8.4.2 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.
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0,>=0.16.4->tokenizers>=0.13.2->chromadb) (8.5.0)
Requirement already satisfied: filelock>=3.10.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.0,>
=0.16.4->tokenizers>=0.13.2->chromadb) (3.32.7)
Requirement already satisfied: fsspec>=2023.5.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.0,>
=0.16.4->tokenizers>=0.13.2->chromadb) (2026.9.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.6.0 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<
2.0,>=0.16.4->tokenizers>=0.13.2->chromadb) (1.6.0)
Requirement already satisfied: shellingham>=1.3.0 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from typer>=0.9.0->chrom
adb) (1.5.4)
Requirement already satisfied: annotated-doc>=0.0.2 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from typer>=0.9.0->ch
romadb) (0.0.5)
Requirement already satisfied: httptools>=0.6.3 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from uvicorn[standard]>=0.
18.3->chromadb) (0.8.0)
Requirement already satisfied: watchfiles>=0.13 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from uvicorn[standard]>=0.
18.3->chromadb) (1.3.0)
Requirement already satisfied: websockets>=10.4 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from uvicorn[standard]>=0.
18.3->chromadb) (16.1.1)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from requests->kub
ernetes>=28.1.0->chromadb) (3.5.1)
Requirement already satisfied: oauthlib>=3.0.0 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from requests-oauthlib->kub
ernetes>=28.1.0->chromadb) (4.0.0)
Using cached protobuf-7.36.2-cp310-abi3-win_amd64.whl (456 kB)
Installing collected packages: protobuf
  Attempting uninstall: protobuf
    Found existing installation: protobuf 4.25.9
    Uninstalling protobuf-4.25.9:
      Successfully uninstalled protobuf-4.25.9
Successfully installed protobuf-7.36.2
ERROR: pip's dependency resolver does not currently take into account all the pac
kages that are installed. This behaviour is the source of the following dependenc
y conflicts.
google-ai-generativelanguage 0.6.6 requires protobuf!=3.20.0,!=3.20.1,!=4.21.0,!=
4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<5.0.0dev,>=3.19.5, but you have proto
buf 7.36.2 which is incompatible.
```

```
Requirement already satisfied: faiss-cpu in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (1.15.1)
Requirement already satisfied: numpy>=1.25 in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from faiss-cpu) (2.4.4)
Requirement already satisfied: packaging in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from faiss-cpu) (25.0)
Requirement already satisfied: google-genai in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (2.25.0)
Requirement already satisfied: anyio<5.0.0,>=4.8.0 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from google-genai) (4.1
5.1)
Requirement already satisfied: google-auth<3.0.0,>=2.56.0 in C:\Users\shreya.2.ag
rawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-auth
[requests]<3.0.0,>=2.56.0->google-genai) (2.58.1)
Requirement already satisfied: httpx<1.0.0,>=0.28.1 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-genai)
(0.28.1)
Requirement already satisfied: pydantic<3.0.0,>=2.12.5 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-genai)
(2.12.5)
Requirement already satisfied: requests<3.0.0,>=2.28.1 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-genai)
(2.34.2)
Requirement already satisfied: tenacity<9.2.0,>=8.2.3 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-genai)
(9.1.4)
Requirement already satisfied: websockets<17.0,>=13.0.0 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-genai)
(16.1.1)
Requirement already satisfied: typing-extensions<5.0.0,>=4.14.0 in C:\Users\shrey
a.2.agrawal\AppData\Roaming\Python\Python313\site-packages (from google-genai)
(4.16.0)
Requirement already satisfied: distro<2,>=1.7.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from google-genai) (1.9.0)
Requirement already satisfied: sniffio in C:\Users\shreya.2.agrawal\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from google-genai) (1.3.1)
Requirement already satisfied: idna>=2.8 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from anyio<5.0.0,>=4.8.0->google-
genai) (3.20)
Requirement already satisfied: pyasn1-modules>=0.2.1 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-auth<3.0.
0,>=2.56.0->google-auth[requests]<3.0.0,>=2.56.0->google-genai) (0.4.2)
Requirement already satisfied: cryptography>=38.0.3 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-auth<3.0.
0,>=2.56.0->google-auth[requests]<3.0.0,>=2.56.0->google-genai) (50.0.1)
Requirement already satisfied: certifi in C:\Users\shreya.2.agrawal\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from httpx<1.0.0,>=0.28.1->google-g
enai) (2026.7.22)
Requirement already satisfied: httpcore==1.* in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from httpx<1.0.0,>=0.28.1->go
ogle-genai) (1.0.9)
Requirement already satisfied: h11>=0.16 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from httpcore==1.*->httpx<1.0.0,>
=0.28.1->google-genai) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3.0.0,
>=2.12.5->google-genai) (0.8.0)
Requirement already satisfied: pydantic-core==2.41.5 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3.0.0,>
=2.12.5->google-genai) (2.41.5)
```

```
Requirement already satisfied: typing-inspection>=0.4.2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3.0.
0,>=2.12.5->google-genai) (0.4.4)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from requests<3.0.
0,>=2.28.1->google-genai) (3.5.1)
Requirement already satisfied: urllib3<3,>=1.26 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from requests<3.0.0,>=2.2
8.1->google-genai) (2.8.0)
Requirement already satisfied: cffi>=2.0.0 in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from cryptography>=38.0.3->goog
le-auth<3.0.0,>=2.56.0->google-auth[requests]<3.0.0,>=2.56.0->google-genai) (2.1.
1)
Requirement already satisfied: pycparser in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from cffi>=2.0.0->cryptography>=3
8.0.3->google-auth<3.0.0,>=2.56.0->google-auth[requests]<3.0.0,>=2.56.0->google-g
enai) (3.0)
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pyasn1-modules>=
0.2.1->google-auth<3.0.0,>=2.56.0->google-auth[requests]<3.0.0,>=2.56.0->google-g
enai) (0.6.4)
```

In [2]:
```
!pip install jq
```

```
Requirement already satisfied: jq in C:\Users\shreya.2.agrawal\AppData\Local\Prog
rams\Python\Python313\Lib\site-packages (1.12.0)
```

In [3]:
```
from langchain.document_loaders import CSVLoader, JSONLoader
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain.embeddings import HuggingFaceEmbeddings
from langchain.vectorstores import Chroma, FAISS
from langchain.chains import RetrievalQA
```

In [4]:
```
jq_schema = ".[] | {instruction: .instruction, input: .input, output: .output}"


csv_docs = CSVLoader(
    file_path="C:/Users/shreya.2.agrawal/Downloads/Medical/Medical/format_datase
).load()

json_docs = JSONLoader(
    file_path="C:/Users/shreya.2.agrawal/Downloads/Medical/Medical/chatdoctor5k.
    jq_schema=jq_schema,
    text_content=False
).load()

documents = csv_docs + json_docs
```

In [5]:
```
recursive_splitter = RecursiveCharacterTextSplitter(chunk_size=500,
                                                    chunk_overlap=100,
                                                    separators=["\n\n", "\n", ".

recursive_tokens = recursive_splitter.split_documents(documents)
```

In [6]:
```
char_splitter = RecursiveCharacterTextSplitter(chunk_size=500,
                                               chunk_overlap=100,
                                               separators=["\n\n", "\n", ".
```

```
recursive_tokens = recursive_splitter.split_documents(documents)
```

In [7]:
```
char_chunks = char_splitter.split_documents(documents)
```

In [8]:
```
char_chunks[0]
```

Out[8]:
```
Document(metadata={'source': 'C:/Users/shreya.2.agrawal/Downloads/Medical/Medic
al/format_dataset.csv', 'row': 0}, page_content="idx: 0\ndisease: Panic disorde
r\nSymptom: ['Anxiety and nervousness', 'Depression', 'Shortness of breath', 'D
epressive or psychotic symptoms', 'Sharp chest pain', 'Dizziness', 'Insomnia',
'Abnormal involuntary movements', 'Chest tightness', 'Palpitations', 'Irregular
heartbeat', 'Breathing fast']")
```

In [9]:
```
char_chunks = char_splitter.split_documents(documents)

char_chunks[0]

len(char_chunks)
```

Out[9]:
```
10964
```

In [10]:
```
# size of the tokens
print(f"Number of tokens created from document: {len(recursive_splitter.split_do
```

```
Number of tokens created from document: 295
```

In [11]:
```
# size of the tokens
print(f"Number of tokens created from document: {len(recursive_tokens)}")
```

```
Number of tokens created from document: 10964
```

In [12]:
```
import os
os.environ["OPENAI_API_KEY"] = "sk-proj-lR8Ii4wPLJflx5dgx5VGJ_3oXXS9IZPF-69SK9PZ
```

In [13]:
```
!pip install langchain-openai==0.3.27

!pip install sentence-transformers
```

```
Requirement already satisfied: langchain-openai==0.3.27 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (0.3.27)
Requirement already satisfied: langchain-core<1.0.0,>=0.3.66 in C:\Users\shreya.
2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langcha
in-openai==0.3.27) (0.3.72)
Requirement already satisfied: openai<2.0.0,>=1.86.0 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-openai
==0.3.27) (1.109.1)
Requirement already satisfied: tiktoken<1,>=0.7 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from langchain-openai==0.
3.27) (0.14.0)
Requirement already satisfied: langsmith>=0.3.45 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<1.0.
0,>=0.3.66->langchain-openai==0.3.27) (0.14.1)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in C:\Users\shrey
a.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langc
hain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (9.1.4)
Requirement already satisfied: jsonpatch<2.0,>=1.33 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<
1.0.0,>=0.3.66->langchain-openai==0.3.27) (1.33)
Requirement already satisfied: PyYAML>=5.3 in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from langchain-core<1.0.0,>=0.
3.66->langchain-openai==0.3.27) (6.0.3)
Requirement already satisfied: typing-extensions>=4.7 in C:\Users\shreya.2.agrawa
l\AppData\Roaming\Python\Python313\site-packages (from langchain-core<1.0.0,>=0.
3.66->langchain-openai==0.3.27) (4.16.0)
Requirement already satisfied: packaging>=23.2 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<1.0.0,>
=0.3.66->langchain-openai==0.3.27) (25.0)
Requirement already satisfied: pydantic>=2.7.4 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<1.0.0,>
=0.3.66->langchain-openai==0.3.27) (2.12.5)
Requirement already satisfied: jsonpointer>=1.9 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from jsonpatch<2.0,>=1.33-
>langchain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (3.1.1)
Requirement already satisfied: anyio<5,>=3.5.0 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from openai<2.0.0,>=1.86.0-
>langchain-openai==0.3.27) (4.15.1)
Requirement already satisfied: distro<2,>=1.7.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from openai<2.0.0,>=1.86.0
->langchain-openai==0.3.27) (1.9.0)
Requirement already satisfied: httpx<1,>=0.23.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from openai<2.0.0,>=1.86.0
->langchain-openai==0.3.27) (0.28.1)
Requirement already satisfied: jiter<1,>=0.4.0 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from openai<2.0.0,>=1.86.0-
>langchain-openai==0.3.27) (0.17.0)
Requirement already satisfied: sniffio in C:\Users\shreya.2.agrawal\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from openai<2.0.0,>=1.86.0->langcha
in-openai==0.3.27) (1.3.1)
Requirement already satisfied: tqdm>4 in C:\Users\shreya.2.agrawal\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from openai<2.0.0,>=1.86.0->langcha
in-openai==0.3.27) (4.70.1)
Requirement already satisfied: idna>=2.8 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from anyio<5,>=3.5.0->openai<2.0.
0,>=1.86.0->langchain-openai==0.3.27) (3.20)
Requirement already satisfied: certifi in C:\Users\shreya.2.agrawal\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from httpx<1,>=0.23.0->openai<2.0.
0,>=1.86.0->langchain-openai==0.3.27) (2026.7.22)
Requirement already satisfied: httpcore==1.* in C:\Users\shreya.2.agrawal\AppData
```

```
\Local\Programs\Python\Python313\Lib\site-packages (from httpx<1,>=0.23.0->openai
<2.0.0,>=1.86.0->langchain-openai==0.3.27) (1.0.9)
Requirement already satisfied: h11>=0.16 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from httpcore==1.*->httpx<1,>=0.2
3.0->openai<2.0.0,>=1.86.0->langchain-openai==0.3.27) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic>=2.7.4
->langchain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (0.8.0)
Requirement already satisfied: pydantic-core==2.41.5 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic>=2.7.4-
>langchain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic>=2.
7.4->langchain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (0.4.4)
Requirement already satisfied: regex in C:\Users\shreya.2.agrawal\AppData\Local\P
rograms\Python\Python313\Lib\site-packages (from tiktoken<1,>=0.7->langchain-open
ai==0.3.27) (2026.9.29)
Requirement already satisfied: requests in C:\Users\shreya.2.agrawal\AppData\Loca
l\Programs\Python\Python313\Lib\site-packages (from tiktoken<1,>=0.7->langchain-o
penai==0.3.27) (2.34.2)
Requirement already satisfied: httpx2<3,>=2 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (2.13.1)
Requirement already satisfied: orjson>=3.9.14 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->lang
chain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (3.12.0)
Requirement already satisfied: requests-toolbelt>=1.0.0 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.
3.45->langchain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (1.0.0)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.
3.45->langchain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (0.17.1)
Requirement already satisfied: websockets>=15.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->la
ngchain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (16.1.1)
Requirement already satisfied: xxhash>=3.0.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (4.0.1)
Requirement already satisfied: zstandard>=0.23.0 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->l
angchain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (0.25.0)
Requirement already satisfied: httpcore2==2.13.1 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsm
ith>=0.3.45->langchain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (2.13.1)
Requirement already satisfied: truststore>=0.10 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsmi
th>=0.3.45->langchain-core<1.0.0,>=0.3.66->langchain-openai==0.3.27) (0.10.4)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from requests->tik
token<1,>=0.7->langchain-openai==0.3.27) (3.5.1)
Requirement already satisfied: urllib3<3,>=1.26 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from requests->tiktoken<1,
>=0.7->langchain-openai==0.3.27) (2.8.0)
Requirement already satisfied: colorama in C:\Users\shreya.2.agrawal\AppData\Roam
ing\Python\Python313\site-packages (from tqdm>4->openai<2.0.0,>=1.86.0->langchain
-openai==0.3.27) (0.4.6)
Requirement already satisfied: sentence-transformers in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (6.1.0)
Requirement already satisfied: transformers<6.0.0,>=5.0.0 in C:\Users\shreya.2.ag
rawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from sentence-tr
```

```
ansformers) (5.17.0)
Requirement already satisfied: tokenizers>=0.19 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from sentence-transformer
s) (0.23.2)
Requirement already satisfied: huggingface-hub<2.0.0,>=1.3.0 in C:\Users\shreya.
2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from sentenc
e-transformers) (1.33.0)
Requirement already satisfied: torch>=2.2 in C:\Users\shreya.2.agrawal\AppData\Lo
cal\Programs\Python\Python313\Lib\site-packages (from sentence-transformers) (2.1
4.0)
Requirement already satisfied: numpy>=1.24.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from sentence-transformers)
(2.4.4)
Requirement already satisfied: scikit-learn>=1.1.0 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from sentence-transform
ers) (1.8.0)
Requirement already satisfied: scipy>=1.0.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from sentence-transformers)
(1.18.1)
Requirement already satisfied: typing_extensions>=4.10.0 in C:\Users\shreya.2.agr
awal\AppData\Roaming\Python\Python313\site-packages (from sentence-transformers)
(4.16.0)
Requirement already satisfied: tqdm>=4.0.0 in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from sentence-transformers) (4.
70.1)
Requirement already satisfied: click<9.0.0,>=8.4.2 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.
0.0,>=1.3.0->sentence-transformers) (8.5.0)
Requirement already satisfied: filelock>=3.10.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.0.
0,>=1.3.0->sentence-transformers) (3.32.7)
Requirement already satisfied: fsspec>=2023.5.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.0.
0,>=1.3.0->sentence-transformers) (2026.9.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.6.0 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<
2.0.0,>=1.3.0->sentence-transformers) (1.6.0)
Requirement already satisfied: httpx<1,>=0.23.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.0.
0,>=1.3.0->sentence-transformers) (0.28.1)
Requirement already satisfied: packaging>=20.9 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.0.0,
>=1.3.0->sentence-transformers) (25.0)
Requirement already satisfied: pyyaml>=5.1 in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.0.0,>=1.
3.0->sentence-transformers) (6.0.3)
Requirement already satisfied: anyio in C:\Users\shreya.2.agrawal\AppData\Local\P
rograms\Python\Python313\Lib\site-packages (from httpx<1,>=0.23.0->huggingface-hu
b<2.0.0,>=1.3.0->sentence-transformers) (4.15.1)
Requirement already satisfied: certifi in C:\Users\shreya.2.agrawal\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from httpx<1,>=0.23.0->huggingface-
hub<2.0.0,>=1.3.0->sentence-transformers) (2026.7.22)
Requirement already satisfied: httpcore==1.* in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from httpx<1,>=0.23.0->huggin
gface-hub<2.0.0,>=1.3.0->sentence-transformers) (1.0.9)
Requirement already satisfied: idna in C:\Users\shreya.2.agrawal\AppData\Local\Pr
ograms\Python\Python313\Lib\site-packages (from httpx<1,>=0.23.0->huggingface-hub
<2.0.0,>=1.3.0->sentence-transformers) (3.20)
Requirement already satisfied: h11>=0.16 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from httpcore==1.*->httpx<1,>=0.2
```

```
3.0->huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (0.16.0)
Requirement already satisfied: regex>=2025.10.22 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from transformers<6.0.0,>
=5.0.0->sentence-transformers) (2026.9.29)
Requirement already satisfied: typer in C:\Users\shreya.2.agrawal\AppData\Local\P
rograms\Python\Python313\Lib\site-packages (from transformers<6.0.0,>=5.0.0->sent
ence-transformers) (0.27.2)
Requirement already satisfied: safetensors>=0.8.0 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from transformers<6.0.0,
>=5.0.0->sentence-transformers) (0.8.0)
Requirement already satisfied: joblib>=1.3.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from scikit-learn>=1.1.0->sen
tence-transformers) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.2.0 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from scikit-learn>=1.
1.0->sentence-transformers) (3.7.0)
Requirement already satisfied: setuptools>=77.0.3 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from torch>=2.2->sentenc
e-transformers) (84.0.0)
Requirement already satisfied: sympy>=1.13.3 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from torch>=2.2->sentence-tra
nsformers) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from torch>=2.2->sentence-t
ransformers) (3.7)
Requirement already satisfied: jinja2 in C:\Users\shreya.2.agrawal\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from torch>=2.2->sentence-transform
ers) (3.1.6)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from sympy>=1.13.3->torc
h>=2.2->sentence-transformers) (1.3.0)
Requirement already satisfied: colorama in C:\Users\shreya.2.agrawal\AppData\Roam
ing\Python\Python313\site-packages (from tqdm>=4.0.0->sentence-transformers) (0.
4.6)
Requirement already satisfied: MarkupSafe>=2.0 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from jinja2->torch>=2.2->se
ntence-transformers) (3.0.3)
Requirement already satisfied: shellingham>=1.3.0 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from typer->transformers
<6.0.0,>=5.0.0->sentence-transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from typer->transformers<6.0.
0,>=5.0.0->sentence-transformers) (15.0.0)
Requirement already satisfied: annotated-doc>=0.0.2 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from typer->transform
ers<6.0.0,>=5.0.0->sentence-transformers) (0.0.5)
Requirement already satisfied: markdown-it-py>=2.2.0 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from rich>=13.8.0->ty
per->transformers<6.0.0,>=5.0.0->sentence-transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in C:\Users\shreya.2.agraw
al\AppData\Roaming\Python\Python313\site-packages (from rich>=13.8.0->typer->tran
sformers<6.0.0,>=5.0.0->sentence-transformers) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in C:\Users\shreya.2.agrawal\AppData\Lo
cal\Programs\Python\Python313\Lib\site-packages (from markdown-it-py>=2.2.0->rich
>=13.8.0->typer->transformers<6.0.0,>=5.0.0->sentence-transformers) (0.1.2)
```

In [14]:
```
from langchain_openai import OpenAIEmbeddings
```

In [21]:
```
openai_embeddings = OpenAIEmbeddings(model = "text-embedding-ada-002",
```

```
                                        openai_api_key = os.environ["OPENAI_API_KEY
```

In [16]:
```
openai_embedding_result = openai_embeddings.embed_documents(documents[0].page_co
```

```
---------------------------------------------------------------------------
ConnectError                              Traceback (most recent call last)
File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_transport
s\default.py:101, in map_httpcore_exceptions()
    100 try:
--> 101     yield
    102 except Exception as exc:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_transport
s\default.py:250, in HTTPTransport.handle_request(self, request)
    249 with map_httpcore_exceptions():
--> 250     resp = self._pool.handle_request(req)
    252 assert isinstance(resp.stream, typing.Iterable)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_sync\c
onnection_pool.py:256, in ConnectionPool.handle_request(self, request)
    255     self._close_connections(closing)
--> 256     raise exc from None
    258 # Return the response. Note that in this case we still have to manage
    259 # the point at which the response is closed.

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_sync\c
onnection_pool.py:236, in ConnectionPool.handle_request(self, request)
    234 try:
    235     # Send the request on the assigned connection.
--> 236     response = connection.handle_request(
    237         pool_request.request
    238     )
    239 except ConnectionNotAvailable:
    240     # In some cases a connection may initially be available to
    241     # handle a request, but then become unavailable.
    242     #
    243     # In this case we clear the connection and try again.

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_sync\c
onnection.py:101, in HTTPConnection.handle_request(self, request)
    100     self._connect_failed = True
--> 101     raise exc
    103 return self._connection.handle_request(request)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_sync\c
onnection.py:78, in HTTPConnection.handle_request(self, request)
     77 if self._connection is None:
---> 78     stream = self._connect(request)
     80     ssl_object = stream.get_extra_info("ssl_object")

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_sync\c
onnection.py:156, in HTTPConnection._connect(self, request)
    155 with Trace("start_tls", logger, request, kwargs) as trace:
--> 156     stream = stream.start_tls(**kwargs)
    157     trace.return_value = stream

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_backen
ds\sync.py:154, in SyncStream.start_tls(self, ssl_context, server_hostname, timeo
ut)
    150 exc_map: ExceptionMapping = {
    151     socket.timeout: ConnectTimeout,
    152     OSError: ConnectError,
    153 }
--> 154 with map_exceptions(exc_map):
```

```
    155     try:

File ~\AppData\Local\Programs\Python\Python313\Lib\contextlib.py:162, in _Generat
orContextManager.__exit__(self, typ, value, traceback)
    161 try:
--> 162     self.gen.throw(value)
    163 except StopIteration as exc:
    164     # Suppress StopIteration *unless* it's the same exception that
    165     # was passed to throw().  This prevents a StopIteration
    166     # raised inside the "with" statement from being suppressed.

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_except
ions.py:14, in map_exceptions(map)
     13     if isinstance(exc, from_exc):
---> 14         raise to_exc(exc) from exc
     15 raise

ConnectError: [SSL: WRONG_VERSION_NUMBER] wrong version number (_ssl.c:1032)

The above exception was the direct cause of the following exception:

ConnectError                              Traceback (most recent call last)
File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\openai\_base_cli
ent.py:982, in SyncAPIClient.request(self, cast_to, options, stream, stream_cls)
    981 try:
--> 982     response = self._client.send(
    983         request,
    984         stream=stream or self._should_stream_response_body(request=reques
t),
    985         **kwargs,
    986     )
    987 except httpx.TimeoutException as err:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_client.p
y:914, in Client.send(self, request, stream, auth, follow_redirects)
    912 auth = self._build_request_auth(request, auth)
--> 914 response = self._send_handling_auth(
    915     request,
    916     auth=auth,
    917     follow_redirects=follow_redirects,
    918     history=[],
    919 )
    920 try:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_client.p
y:942, in Client._send_handling_auth(self, request, auth, follow_redirects, histo
ry)
    941 while True:
--> 942     response = self._send_handling_redirects(
    943         request,
    944         follow_redirects=follow_redirects,
    945         history=history,
    946     )
    947     try:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_client.p
y:979, in Client._send_handling_redirects(self, request, follow_redirects, histor
y)
    977     hook(request)
--> 979 response = self._send_single_request(request)
```

```
    980 try:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_client.py:1014, in Client._send_single_request(self, request)
   1013 with request_context(request=request):
-> 1014     response = transport.handle_request(request)
   1016 assert isinstance(response.stream, SyncByteStream)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_transports\default.py:249, in HTTPTransport.handle_request(self, request)
    237 req = httpcore.Request(
    238     method=request.method,
    239     url=httpcore.URL(
   (...)    247     extensions=request.extensions,
    248 )
--> 249 with map_httpcore_exceptions():
    250     resp = self._pool.handle_request(req)

File ~\AppData\Local\Programs\Python\Python313\Lib\contextlib.py:162, in _GeneratorContextManager.__exit__(self, typ, value, traceback)
    161 try:
--> 162     self.gen.throw(value)
    163 except StopIteration as exc:
    164     # Suppress StopIteration *unless* it's the same exception that
    165     # was passed to throw().  This prevents a StopIteration
    166     # raised inside the "with" statement from being suppressed.

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_transports\default.py:118, in map_httpcore_exceptions()
    117 message = str(exc)
--> 118 raise mapped_exc(message) from exc

ConnectError: [SSL: WRONG_VERSION_NUMBER] wrong version number (_ssl.c:1032)

The above exception was the direct cause of the following exception:

APIConnectionError                        Traceback (most recent call last)
Cell In[16], line 1
----> 1 openai_embedding_result = openai_embeddings.embed_documents(documents[0].page_content)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\langchain_openai\embeddings\base.py:590, in OpenAIEmbeddings.embed_documents(self, texts, chunk_size, **kwargs)
    587 # NOTE: to keep things simple, we assume the list may contain texts longer
    588 #       than the maximum context and use length-safe embedding function.
    589 engine = cast(str, self.deployment)
--> 590 return self._get_len_safe_embeddings(
    591     texts, engine=engine, chunk_size=chunk_size, **kwargs
    592 )

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\langchain_openai\embeddings\base.py:478, in OpenAIEmbeddings._get_len_safe_embeddings(self, texts, engine, chunk_size, **kwargs)
    476 batched_embeddings: list[list[float]] = []
    477 for i in _iter:
--> 478     response = self.client.create(
    479         input=tokens[i : i + _chunk_size], **client_kwargs
    480     )
```

```
    481     if not isinstance(response, dict):
    482         response = response.model_dump()

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\openai\resources
\embeddings.py:132, in Embeddings.create(self, input, model, dimensions, encoding
_format, user, extra_headers, extra_query, extra_body, timeout)
    126             embedding.embedding = np.frombuffer(  # type: ignore[no-untyp
ed-call]
    127                 base64.b64decode(data), dtype="float32"
    128             ).tolist()
    130     return obj
--> 132 return self._post(
    133     "/embeddings",
    134     body=maybe_transform(params, embedding_create_params.EmbeddingCreateP
arams),
    135     options=make_request_options(
    136         extra_headers=extra_headers,
    137         extra_query=extra_query,
    138         extra_body=extra_body,
    139         timeout=timeout,
    140         post_parser=parser,
    141     ),
    142     cast_to=CreateEmbeddingResponse,
    143 )

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\openai\_base_cli
ent.py:1259, in SyncAPIClient.post(self, path, cast_to, body, options, files, str
eam, stream_cls)
   1245 def post(
   1246     self,
   1247     path: str,
   (...)   1254     stream_cls: type[_StreamT] | None = None,
   1255 ) -> ResponseT | _StreamT:
   1256     opts = FinalRequestOptions.construct(
   1257         method="post", url=path, json_data=body, files=to_httpx_files(fil
es), **options
   1258     )
-> 1259     return cast(ResponseT, self.request(cast_to, opts, stream=stream, str
eam_cls=stream_cls))

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\openai\_base_cli
ent.py:1014, in SyncAPIClient.request(self, cast_to, options, stream, stream_cls)
   1011         continue
   1013     log.debug("Raising connection error")
-> 1014     raise APIConnectionError(request=request) from err
   1016 log.debug(
   1017     'HTTP Response: %s %s "%i %s" %s',
   1018     request.method,
   (...)   1022     response.headers,
   1023 )
   1024 log.debug("request_id: %s", response.headers.get("x-request-id"))

APIConnectionError: Connection error.
```

In [17]:
```
openai_embedding_result
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[17], line 1
----> 1 openai_embedding_result

NameError: name 'openai_embedding_result' is not defined
```

In [22]:
```
from langchain_community.vectorstores import FAISS
```

In [27]:
```
faiss_store = FAISS.from_documents(documents = recursive_tokens,
                                   embedding= openai_embeddings)
```

```
---------------------------------------------------------------------------
ConnectError                              Traceback (most recent call last)
File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_transport
s\default.py:101, in map_httpcore_exceptions()
    100 try:
--> 101     yield
    102 except Exception as exc:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_transport
s\default.py:250, in HTTPTransport.handle_request(self, request)
    249 with map_httpcore_exceptions():
--> 250     resp = self._pool.handle_request(req)
    252 assert isinstance(resp.stream, typing.Iterable)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_sync\c
onnection_pool.py:256, in ConnectionPool.handle_request(self, request)
    255     self._close_connections(closing)
--> 256     raise exc from None
    258 # Return the response. Note that in this case we still have to manage
    259 # the point at which the response is closed.

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_sync\c
onnection_pool.py:236, in ConnectionPool.handle_request(self, request)
    234 try:
    235     # Send the request on the assigned connection.
--> 236     response = connection.handle_request(
    237         pool_request.request
    238     )
    239 except ConnectionNotAvailable:
    240     # In some cases a connection may initially be available to
    241     # handle a request, but then become unavailable.
    242     #
    243     # In this case we clear the connection and try again.

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_sync\c
onnection.py:101, in HTTPConnection.handle_request(self, request)
    100     self._connect_failed = True
--> 101     raise exc
    103 return self._connection.handle_request(request)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_sync\c
onnection.py:78, in HTTPConnection.handle_request(self, request)
     77 if self._connection is None:
---> 78     stream = self._connect(request)
     80     ssl_object = stream.get_extra_info("ssl_object")

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_sync\c
onnection.py:156, in HTTPConnection._connect(self, request)
    155 with Trace("start_tls", logger, request, kwargs) as trace:
--> 156     stream = stream.start_tls(**kwargs)
    157     trace.return_value = stream

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_backen
ds\sync.py:154, in SyncStream.start_tls(self, ssl_context, server_hostname, timeo
ut)
    150 exc_map: ExceptionMapping = {
    151     socket.timeout: ConnectTimeout,
    152     OSError: ConnectError,
    153 }
--> 154 with map_exceptions(exc_map):
```

```
    155     try:

File ~\AppData\Local\Programs\Python\Python313\Lib\contextlib.py:162, in _Generat
orContextManager.__exit__(self, typ, value, traceback)
    161 try:
--> 162     self.gen.throw(value)
    163 except StopIteration as exc:
    164     # Suppress StopIteration *unless* it's the same exception that
    165     # was passed to throw().  This prevents a StopIteration
    166     # raised inside the "with" statement from being suppressed.

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpcore\_except
ions.py:14, in map_exceptions(map)
     13     if isinstance(exc, from_exc):
---> 14         raise to_exc(exc) from exc
     15 raise

ConnectError: [SSL: WRONG_VERSION_NUMBER] wrong version number (_ssl.c:1032)

The above exception was the direct cause of the following exception:

ConnectError                              Traceback (most recent call last)
File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\openai\_base_cli
ent.py:982, in SyncAPIClient.request(self, cast_to, options, stream, stream_cls)
    981 try:
--> 982     response = self._client.send(
    983         request,
    984         stream=stream or self._should_stream_response_body(request=reques
t),
    985         **kwargs,
    986     )
    987 except httpx.TimeoutException as err:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_client.p
y:914, in Client.send(self, request, stream, auth, follow_redirects)
    912 auth = self._build_request_auth(request, auth)
--> 914 response = self._send_handling_auth(
    915     request,
    916     auth=auth,
    917     follow_redirects=follow_redirects,
    918     history=[],
    919 )
    920 try:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_client.p
y:942, in Client._send_handling_auth(self, request, auth, follow_redirects, histo
ry)
    941 while True:
--> 942     response = self._send_handling_redirects(
    943         request,
    944         follow_redirects=follow_redirects,
    945         history=history,
    946     )
    947     try:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_client.p
y:979, in Client._send_handling_redirects(self, request, follow_redirects, histor
y)
    977     hook(request)
--> 979 response = self._send_single_request(request)
```

```
    980 try:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_client.py:1014, in Client._send_single_request(self, request)
   1013 with request_context(request=request):
-> 1014     response = transport.handle_request(request)
   1016 assert isinstance(response.stream, SyncByteStream)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_transports\default.py:249, in HTTPTransport.handle_request(self, request)
    237 req = httpcore.Request(
    238     method=request.method,
    239     url=httpcore.URL(
   (...)    247     extensions=request.extensions,
    248 )
--> 249 with map_httpcore_exceptions():
    250     resp = self._pool.handle_request(req)

File ~\AppData\Local\Programs\Python\Python313\Lib\contextlib.py:162, in _GeneratorContextManager.__exit__(self, typ, value, traceback)
    161 try:
--> 162     self.gen.throw(value)
    163 except StopIteration as exc:
    164     # Suppress StopIteration *unless* it's the same exception that
    165     # was passed to throw().  This prevents a StopIteration
    166     # raised inside the "with" statement from being suppressed.

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\httpx\_transports\default.py:118, in map_httpcore_exceptions()
    117 message = str(exc)
--> 118 raise mapped_exc(message) from exc

ConnectError: [SSL: WRONG_VERSION_NUMBER] wrong version number (_ssl.c:1032)

The above exception was the direct cause of the following exception:

APIConnectionError                        Traceback (most recent call last)
Cell In[27], line 1
----> 1 faiss_store = FAISS.from_documents(documents = recursive_tokens,
      2                                    embedding= openai_embeddings)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\langchain_core\vectorstores\base.py:848, in VectorStore.from_documents(cls, documents, embedding, **kwargs)
    845     if any(ids):
    846         kwargs["ids"] = ids
--> 848 return cls.from_texts(texts, embedding, metadatas=metadatas, **kwargs)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\langchain_community\vectorstores\faiss.py:1043, in FAISS.from_texts(cls, texts, embedding, metadatas, ids, **kwargs)
   1016 @classmethod
   1017 def from_texts(
   1018     cls,
   (...)   1023     **kwargs: Any,
   1024 ) -> FAISS:
   1025     """Construct FAISS wrapper from raw documents.
   1026
   1027     This is a user friendly interface that:
   (...)   1041             faiss = FAISS.from_texts(texts, embeddings)
```

```
   1042     """
-> 1043     embeddings = embedding.embed_documents(texts)
   1044     return cls.__from(
   1045         texts,
   1046         embeddings,
   (...)   1050         **kwargs,
   1051     )

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\langchain_openai
\embeddings\base.py:590, in OpenAIEmbeddings.embed_documents(self, texts, chunk_s
ize, **kwargs)
    587 # NOTE: to keep things simple, we assume the list may contain texts longe
r
    588 #       than the maximum context and use length-safe embedding function.
    589 engine = cast(str, self.deployment)
--> 590 return self._get_len_safe_embeddings(
    591     texts, engine=engine, chunk_size=chunk_size, **kwargs
    592 )

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\langchain_openai
\embeddings\base.py:478, in OpenAIEmbeddings._get_len_safe_embeddings(self, text
s, engine, chunk_size, **kwargs)
    476 batched_embeddings: list[list[float]] = []
    477 for i in _iter:
--> 478     response = self.client.create(
    479         input=tokens[i : i + _chunk_size], **client_kwargs
    480     )
    481     if not isinstance(response, dict):
    482         response = response.model_dump()

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\openai\resources
\embeddings.py:132, in Embeddings.create(self, input, model, dimensions, encoding
_format, user, extra_headers, extra_query, extra_body, timeout)
    126             embedding.embedding = np.frombuffer(  # type: ignore[no-untyp
ed-call]
    127                 base64.b64decode(data), dtype="float32"
    128             ).tolist()
    130     return obj
--> 132 return self._post(
    133     "/embeddings",
    134     body=maybe_transform(params, embedding_create_params.EmbeddingCreateP
arams),
    135     options=make_request_options(
    136         extra_headers=extra_headers,
    137         extra_query=extra_query,
    138         extra_body=extra_body,
    139         timeout=timeout,
    140         post_parser=parser,
    141     ),
    142     cast_to=CreateEmbeddingResponse,
    143 )

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\openai\_base_cli
ent.py:1259, in SyncAPIClient.post(self, path, cast_to, body, options, files, str
eam, stream_cls)
   1245 def post(
   1246     self,
   1247     path: str,
   (...)   1254     stream_cls: type[_StreamT] | None = None,
   1255 ) -> ResponseT | _StreamT:
```

```
   1256     opts = FinalRequestOptions.construct(
   1257         method="post", url=path, json_data=body, files=to_httpx_files(fil
es), **options
   1258     )
-> 1259     return cast(ResponseT, self.request(cast_to, opts, stream=stream, str
eam_cls=stream_cls))

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\openai\_base_cli
ent.py:1014, in SyncAPIClient.request(self, cast_to, options, stream, stream_cls)
   1011         continue
   1013     log.debug("Raising connection error")
-> 1014     raise APIConnectionError(request=request) from err
   1016 log.debug(
   1017     'HTTP Response: %s %s "%i %s" %s',
   1018     request.method,
   (...)   1022     response.headers,
   1023 )
   1024 log.debug("request_id: %s", response.headers.get("x-request-id"))

APIConnectionError: Connection error.
```

In [29]:
```
faiss_store.save_local("./chatdoc_faiss_db")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[29], line 1
----> 1 faiss_store.save_local("./chatdoc_faiss_db")

NameError: name 'faiss_store' is not defined
```

In [30]:
```
faiss_store = FAISS.load_local("./chatdoc_faiss_db", embeddings = openai_embeddi
                               allow_dangerous_deserialization = True)
```

```
---------------------------------------------------------------------------
RuntimeError                              Traceback (most recent call last)
Cell In[30], line 1
----> 1 faiss_store = FAISS.load_local("./chatdoc_faiss_db", embeddings = openai_
embeddings,
      2                                allow_dangerous_deserialization = True)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\langchain_commun
ity\vectorstores\faiss.py:1205, in FAISS.load_local(cls, folder_path, embeddings,
index_name, allow_dangerous_deserialization, **kwargs)
   1203 # load index separately since it is not picklable
   1204 faiss = dependable_faiss_import()
-> 1205 index = faiss.read_index(str(path / f"{index_name}.faiss"))
   1207 # load docstore and index_to_docstore_id
   1208 with open(path / f"{index_name}.pkl", "rb") as f:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\faiss\swigfaiss.
py:13878, in read_index(*args)
  13877 def read_index(*args):
> 13878     return _swigfaiss.read_index(*args)

RuntimeError: Error in __cdecl faiss::FileIOReader::FileIOReader(const char *) at
D:\a\faiss\faiss\faiss\impl\io.cpp:70: Error: 'f' failed: could not open chatdoc_
faiss_db\index.faiss for reading: No such file or directory
```

In [32]:
```
!pip install langchain-cohere==0.3.4
!pip install rank_bm25
```

```
Collecting langchain-cohere==0.3.4
  Downloading langchain_cohere-0.3.4-py3-none-any.whl.metadata (6.7 kB)
Requirement already satisfied: cohere<6.0,>=5.5.6 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-cohere==
0.3.4) (5.21.1)
Requirement already satisfied: langchain-core<0.4.0,>=0.3.27 in C:\Users\shreya.
2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langcha
in-cohere==0.3.4) (0.3.72)
Collecting langchain-experimental<0.4.0,>=0.3.0 (from langchain-cohere==0.3.4)
  Downloading langchain_experimental-0.3.4-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: pandas>=1.4.3 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langchain-cohere==0.3.4)
(3.0.3)
Requirement already satisfied: pydantic<3,>=2 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from langchain-cohere==0.3.
4) (2.12.5)
Collecting tabulate<0.10.0,>=0.9.0 (from langchain-cohere==0.3.4)
  Downloading tabulate-0.9.0-py3-none-any.whl.metadata (34 kB)
Requirement already satisfied: fastavro<2.0.0,>=1.9.4 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from cohere<6.0,>=5.
5.6->langchain-cohere==0.3.4) (1.12.2)
Requirement already satisfied: httpx>=0.21.2 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from cohere<6.0,>=5.5.6->lang
chain-cohere==0.3.4) (0.28.1)
Requirement already satisfied: pydantic-core>=2.18.2 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from cohere<6.0,>=5.
5.6->langchain-cohere==0.3.4) (2.41.5)
Requirement already satisfied: requests<3.0.0,>=2.0.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from cohere<6.0,>=5.
5.6->langchain-cohere==0.3.4) (2.34.2)
Requirement already satisfied: tokenizers<1,>=0.15 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from cohere<6.0,>=5.5.6
->langchain-cohere==0.3.4) (0.23.2)
Requirement already satisfied: types-requests<3.0.0,>=2.0.0 in C:\Users\shreya.2.
agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from cohere<6.
0,>=5.5.6->langchain-cohere==0.3.4) (2.33.0.20260906)
Requirement already satisfied: typing_extensions>=4.0.0 in C:\Users\shreya.2.agra
wal\AppData\Roaming\Python\Python313\site-packages (from cohere<6.0,>=5.5.6->lang
chain-cohere==0.3.4) (4.16.0)
Requirement already satisfied: langsmith>=0.3.45 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<0.4.
0,>=0.3.27->langchain-cohere==0.3.4) (0.14.1)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in C:\Users\shrey
a.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langc
hain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (9.1.4)
Requirement already satisfied: jsonpatch<2.0,>=1.33 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<
0.4.0,>=0.3.27->langchain-cohere==0.3.4) (1.33)
Requirement already satisfied: PyYAML>=5.3 in C:\Users\shreya.2.agrawal\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from langchain-core<0.4.0,>=0.
3.27->langchain-cohere==0.3.4) (6.0.3)
Requirement already satisfied: packaging>=23.2 in C:\Users\shreya.2.agrawal\AppDa
ta\Local\Programs\Python\Python313\Lib\site-packages (from langchain-core<0.4.0,>
=0.3.27->langchain-cohere==0.3.4) (25.0)
Requirement already satisfied: jsonpointer>=1.9 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from jsonpatch<2.0,>=1.33-
>langchain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (3.1.1)
Requirement already satisfied: langchain-community<0.4.0,>=0.3.0 in C:\Users\shre
ya.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from lang
chain-experimental<0.4.0,>=0.3.0->langchain-cohere==0.3.4) (0.3.27)
```

```
Requirement already satisfied: langchain<1.0.0,>=0.3.26 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-com
munity<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langchain-cohere==0.
3.4) (0.3.27)
Requirement already satisfied: SQLAlchemy<3,>=1.4 in C:\Users\shreya.2.agrawal\Ap
pData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-community
<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langchain-cohere==0.3.4)
(2.1.1)
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-commun
ity<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langchain-cohere==0.3.4)
(3.14.3)
Requirement already satisfied: dataclasses-json<0.7,>=0.5.7 in C:\Users\shreya.2.
agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain
-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langchain-cohere=
=0.3.4) (0.6.7)
Requirement already satisfied: pydantic-settings<3.0.0,>=2.4.0 in C:\Users\shrey
a.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langc
hain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langchain-coh
ere==0.3.4) (2.15.0)
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langchain-comm
unity<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langchain-cohere==0.3.
4) (0.4.3)
Requirement already satisfied: numpy>=2.1.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langchain-community<0.4.
0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langchain-cohere==0.3.4) (2.4.4)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,
>=3.8.3->langchain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0-
>langchain-cohere==0.3.4) (2.7.1)
Requirement already satisfied: aiosignal>=1.4.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.8.3
->langchain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langch
ain-cohere==0.3.4) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.8.3->l
angchain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langchain
-cohere==0.3.4) (26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.8.
3->langchain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langc
hain-cohere==0.3.4) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.
8.3->langchain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->lan
gchain-cohere==0.3.4) (6.9.1)
Requirement already satisfied: propcache>=0.2.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.8.3
->langchain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langch
ain-cohere==0.3.4) (0.5.4)
Requirement already satisfied: yarl<2.0,>=1.17.0 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from aiohttp<4.0.0,>=3.8.
3->langchain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,>=0.3.0->langc
hain-cohere==0.3.4) (1.25.1)
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in C:\Users\shreya.2.ag
rawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from dataclasses
-json<0.7,>=0.5.7->langchain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.
0,>=0.3.0->langchain-cohere==0.3.4) (3.26.2)
Requirement already satisfied: typing-inspect<1,>=0.4.0 in C:\Users\shreya.2.agra
```

```
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from dataclasses-j
son<0.7,>=0.5.7->langchain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,
>=0.3.0->langchain-cohere==0.3.4) (0.9.0)
Requirement already satisfied: langchain-text-splitters<1.0.0,>=0.3.9 in C:\Users
\shreya.2.agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
langchain<1.0.0,>=0.3.26->langchain-community<0.4.0,>=0.3.0->langchain-experiment
al<0.4.0,>=0.3.0->langchain-cohere==0.3.4) (0.3.9)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3,>=2-
>langchain-cohere==0.3.4) (0.8.0)
Requirement already satisfied: typing-inspection>=0.4.2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic<3,>=
2->langchain-cohere==0.3.4) (0.4.4)
Requirement already satisfied: python-dotenv>=0.21.0 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from pydantic-setting
s<3.0.0,>=2.4.0->langchain-community<0.4.0,>=0.3.0->langchain-experimental<0.4.0,
>=0.3.0->langchain-cohere==0.3.4) (1.2.3)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from requests<3.0.
0,>=2.0.0->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (3.5.1)
Requirement already satisfied: idna<4,>=2.5 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from requests<3.0.0,>=2.0.0->
cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (3.20)
Requirement already satisfied: urllib3<3,>=1.26 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from requests<3.0.0,>=2.0.
0->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (2.8.0)
Requirement already satisfied: certifi>=2023.5.7 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from requests<3.0.0,>=2.
0.0->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (2026.7.22)
Requirement already satisfied: huggingface-hub<2.0,>=0.16.4 in C:\Users\shreya.2.
agrawal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from tokenizer
s<1,>=0.15->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (1.33.0)
Requirement already satisfied: click<9.0.0,>=8.4.2 in C:\Users\shreya.2.agrawal\A
ppData\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.
0,>=0.16.4->tokenizers<1,>=0.15->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (8.
5.0)
Requirement already satisfied: filelock>=3.10.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.0,>
=0.16.4->tokenizers<1,>=0.15->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (3.32.
7)
Requirement already satisfied: fsspec>=2023.5.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.0,>
=0.16.4->tokenizers<1,>=0.15->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (2026.
9.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.6.0 in C:\Users\shreya.2.agrawal
\AppData\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<
2.0,>=0.16.4->tokenizers<1,>=0.15->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4)
(1.6.0)
Requirement already satisfied: tqdm>=4.42.1 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from huggingface-hub<2.0,>=0.
16.4->tokenizers<1,>=0.15->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (4.70.1)
Requirement already satisfied: anyio in C:\Users\shreya.2.agrawal\AppData\Local\P
rograms\Python\Python313\Lib\site-packages (from httpx>=0.21.2->cohere<6.0,>=5.5.
6->langchain-cohere==0.3.4) (4.15.1)
Requirement already satisfied: httpcore==1.* in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from httpx>=0.21.2->cohere<6.
0,>=5.5.6->langchain-cohere==0.3.4) (1.0.9)
Requirement already satisfied: h11>=0.16 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from httpcore==1.*->httpx>=0.21.2
->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (0.16.0)
```

```
Requirement already satisfied: mypy-extensions>=0.3.0 in C:\Users\shreya.2.agrawa
l\AppData\Local\Programs\Python\Python313\Lib\site-packages (from typing-inspect<
1,>=0.4.0->dataclasses-json<0.7,>=0.5.7->langchain-community<0.4.0,>=0.3.0->langc
hain-experimental<0.4.0,>=0.3.0->langchain-cohere==0.3.4) (1.1.0)
Requirement already satisfied: distro>=1.7.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (1.9.0)
Requirement already satisfied: httpx2<3,>=2 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (2.13.1)
Requirement already satisfied: orjson>=3.9.14 in C:\Users\shreya.2.agrawal\AppDat
a\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->lang
chain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (3.12.0)
Requirement already satisfied: requests-toolbelt>=1.0.0 in C:\Users\shreya.2.agra
wal\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.
3.45->langchain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (1.0.0)
Requirement already satisfied: sniffio>=1.1 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (1.3.1)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in C:\Users\shreya.2.agraw
al\AppData\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.
3.45->langchain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (0.17.1)
Requirement already satisfied: websockets>=15.0 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->la
ngchain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (16.1.1)
Requirement already satisfied: xxhash>=3.0.0 in C:\Users\shreya.2.agrawal\AppData
\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->langc
hain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (4.0.1)
Requirement already satisfied: zstandard>=0.23.0 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from langsmith>=0.3.45->l
angchain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (0.25.0)
Requirement already satisfied: httpcore2==2.13.1 in C:\Users\shreya.2.agrawal\App
Data\Local\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsm
ith>=0.3.45->langchain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (2.13.1)
Requirement already satisfied: truststore>=0.10 in C:\Users\shreya.2.agrawal\AppD
ata\Local\Programs\Python\Python313\Lib\site-packages (from httpx2<3,>=2->langsmi
th>=0.3.45->langchain-core<0.4.0,>=0.3.27->langchain-cohere==0.3.4) (0.10.4)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\shreya.2.agrawa
l\AppData\Roaming\Python\Python313\site-packages (from pandas>=1.4.3->langchain-c
ohere==0.3.4) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\shreya.2.agrawal\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from pandas>=1.4.3->langchain-coher
e==0.3.4) (2026.4)
Requirement already satisfied: six>=1.5 in C:\Users\shreya.2.agrawal\AppData\Roam
ing\Python\Python313\site-packages (from python-dateutil>=2.8.2->pandas>=1.4.3->l
angchain-cohere==0.3.4) (1.17.0)
Requirement already satisfied: colorama in C:\Users\shreya.2.agrawal\AppData\Roam
ing\Python\Python313\site-packages (from tqdm>=4.42.1->huggingface-hub<2.0,>=0.1
6.4->tokenizers<1,>=0.15->cohere<6.0,>=5.5.6->langchain-cohere==0.3.4) (0.4.6)
Downloading langchain_cohere-0.3.4-py3-none-any.whl (44 kB)
Downloading langchain_experimental-0.3.4-py3-none-any.whl (209 kB)
Downloading tabulate-0.9.0-py3-none-any.whl (35 kB)
Installing collected packages: tabulate, langchain-experimental, langchain-cohere

   ------------ -------------------------- 1/3 [langchain-experimental]
   ------------ -------------------------- 1/3 [langchain-experimental]
   ------------ -------------------------- 1/3 [langchain-experimental]
   ------------ -------------------------- 1/3 [langchain-experimental]
   ------------ -------------------------- 1/3 [langchain-experimental]
   ------------ -------------------------- 1/3 [langchain-experimental]
```

```
   ------------ -------------------------- 1/3 [langchain-experimental]
   ------------ -------------------------- 1/3 [langchain-experimental]
  Attempting uninstall: langchain-cohere
   ------------ -------------------------- 1/3 [langchain-experimental]
    Found existing installation: langchain-cohere 0.4.4
   ------------ -------------------------- 1/3 [langchain-experimental]
    Uninstalling langchain-cohere-0.4.4:
   ------------ -------------------------- 1/3 [langchain-experimental]
      Successfully uninstalled langchain-cohere-0.4.4
   ------------ -------------------------- 1/3 [langchain-experimental]
   ------------------------- ------------- 2/3 [langchain-cohere]
   ------------------------- ------------- 2/3 [langchain-cohere]
   ------------------------- ------------- 2/3 [langchain-cohere]
   ---------------------------------------- 3/3 [langchain-cohere]

Successfully installed langchain-cohere-0.3.4 langchain-experimental-0.3.4 tabula
te-0.9.0
Requirement already satisfied: rank_bm25 in C:\Users\shreya.2.agrawal\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (0.2.2)
Requirement already satisfied: numpy in C:\Users\shreya.2.agrawal\AppData\Local\P
rograms\Python\Python313\Lib\site-packages (from rank_bm25) (2.4.4)
```

In [33]:
```
import os
os.environ["OPENAI_API_KEY"] = "sk-proj-il_T9aj8H_sdy7ggXSPAOaAGZJO0JOvf1Yu4113i
```

In [34]:
```
os.environ["COHERE_API_KEY"] = "iodgJ1kyf66POwRf5SR7xBLsuoSygJKryoSywMAo"
```

In [35]:
```
from langchain_openai import ChatOpenAI
llm = ChatOpenAI(model_name="gpt-4.1-nano",
                 temperature=0.7,
                 top_p=0.75,
                 max_tokens=1024,)
```

6. Create Buffer Memory to store Conversational Memory

In [36]:
```
from langchain.memory import ConversationBufferMemory
memory = ConversationBufferMemory(memory_key="chat_history", input_key="question
                                  return_messages=True, output_key='answer')
```

```
C:\Users\shreya.2.agrawal\AppData\Local\Temp\ipykernel_20180\3012205706.py:2: Lan
gChainDeprecationWarning: Please see the migration guide at: https://python.langc
hain.com/docs/versions/migrating_memory/
  memory = ConversationBufferMemory(memory_key="chat_history", input_key="questio
n",
```

In [37]:
```
from langchain_community.retrievers import BM25Retriever
# keyword retriver
bm25_retriever = BM25Retriever.from_documents(documents=recursive_chunks, k=8)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[37], line 3
      1 from langchain_community.retrievers import BM25Retriever
      2 # keyword retriver
----> 3 bm25_retriever = BM25Retriever.from_documents(documents=recursive_chunks,
k=8)

NameError: name 'recursive_chunks' is not defined
```

In [38]:
```
semantic_retriever = faiss_store.as_retriever(search_type="similarity",
                                              search_kwargs={"k": 8})
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[38], line 1
----> 1 semantic_retriever = faiss_store.as_retriever(search_type="similarity",
      2                                               search_kwargs={"k": 8})

NameError: name 'faiss_store' is not defined
```

In [39]:
```
from langchain.retrievers import EnsembleRetriever
# Hybrid Search - Ensemble Retriever
ensemble_retriever = EnsembleRetriever(retrievers=[semantic_retriever,
                                                   bm25_retriever],
                                                   weights=[0.6, 0.4],)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[39], line 3
      1 from langchain.retrievers import EnsembleRetriever
      2 # Hybrid Search - Ensemble Retriever
----> 3 ensemble_retriever = EnsembleRetriever(retrievers=[semantic_retriever,
      4                                                    bm25_retriever],
      5                                                    weights=[0.6, 0.4],)

NameError: name 'semantic_retriever' is not defined
```

In [40]:
```
from langchain.retrievers.contextual_compression import ContextualCompressionRet
```

In [42]:
```
contexual_retriever = ContextualCompressionRetriever(
    base_compressor=reranker,
    base_retriever=ensemble_retriever
)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[42], line 2
      1 contexual_retriever = ContextualCompressionRetriever(
----> 2     base_compressor=reranker,
      3     base_retriever=ensemble_retriever
      4 )

NameError: name 'reranker' is not defined
```

In [43]:
```
prompt = PromptTemplate(
    input_variables=["context", "chat_history", "question"],
    template="""
    You are a medical consulatant with expertise in understanding doctor-patient
    disease symptoms, disease diagnosis, and medical treatments. You have access
    context and chat history. Use only this information to provide a detailed an
    to the patient's question. Stay strictly within the scope of the provided co
    chat history. If the answer is not contained within the context or chat hist
    "I am unable to provide answer."
    Context:
    {context}

    Chat History:
    {chat_history}
```

```
    Question: {question}

    Answer:""")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[43], line 1
----> 1 prompt = PromptTemplate(
      2     input_variables=["context", "chat_history", "question"],
      3     template="""
      4     You are a medical consulatant with expertise in understanding doctor-
patient conversations,

NameError: name 'PromptTemplate' is not defined
```

In [44]:
```
QA_chain = ConversationalRetrievalChain.from_llm(
    llm=llm,
    retriever=contexual_retriever,
    memory=memory, \
    return_source_documents=True,      # to return the source documents along wi
    verbose=False,
    combine_docs_chain_kwargs={"prompt": prompt},
    output_key='answer'
)

print("Welcome to the AI Medical Consultant Chatbot!")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[44], line 1
----> 1 QA_chain = ConversationalRetrievalChain.from_llm(
      2     llm=llm,
      3     retriever=contexual_retriever,
      4     memory=memory, \

NameError: name 'ConversationalRetrievalChain' is not defined
```

In [45]:
```
def ask_question(question: str) -> dict:
    result = QA_chain.invoke({"question": question})

    print("=" * 70)
    print(f"Q: {question}")
    print(f"A: {result['answer']}")
    print("-" * 70)
    print("Sources:")
    for i, doc in enumerate(result["source_documents"], start=1):
        meta   = doc.metadata
        source = meta.get("source", "Unknown").split("\\")[-1]  # filename only
        page   = meta.get("page", "?")
        print(f"  [{i}] {source}  |  Page {page}")
        print(f"       {doc.page_content[:150].strip()}...")
    print("=" * 70)
```

In [46]:
```
ask_question("I am suffering from severe headaches and nausea. What could be the
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[46], line 1
----> 1 ask_question("I am suffering from severe headaches and nausea. What could be the possible causes and what should I do?")

Cell In[45], line 2, in ask_question(question)
      1 def ask_question(question: str) -> dict:
----> 2     result = QA_chain.invoke({"question": question})
      3
      4     print("=" * 70)
      5     print(f"Q: {question}")

NameError: name 'QA_chain' is not defined
```